Group ID	5
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Title	Flux — Visual Data Pipeline Builder
Brief Description	A browser-based visual data pipeline tool that allows users to upload CSV files, build workflows by connecting transformation nodes, preview results, and generate Excel formulas for the pipeline logic.

Serial	FR ID	Functional Requirement	User Story ID	US Title	As a (precise role)	I want (requirement)	So that (completion condition)	User Story	Acceptance Criteria	Priority	Status
1	FR_1	The system shall allow new users to register an account.	US_1	Register account	New User	to register a new account	I can save and access my pipelines	As a New User, I want to register a new account so that I can save and access my pipelines.	User submits a registration form with email and password. System validates input and creates the account. User receives confirmation of successful registration.	Medium	To Do
2	FR_2	The system shall allow registered users to log in with their credentials, email and password.	US_2	Log in	Registered User	to log in with my email and password	I can access my saved pipelines and templates	As a Registered User, I want to log in with my credentials so that I can access my saved pipelines and templates.	User submits email and password. System validates credentials and issues an authentication token. User is redirected to the home page on success.	Medium	To Do
3	FR_3	The system shall display a list of the user's saved pipeline templates.	US_3	Browse templates	Registered User	to view a list of my saved pipeline templates	I can choose one to reuse	As a Registered User, I want to view a list of my saved pipeline templates so that I can choose one to reuse.	System displays all templates owned by the logged-in user. Each entry shows template name and last modified date. User can select a template to load it.	Medium	To Do
4	FR_4	The system shall allow users to import source data by uploading supported file formats, specifically including standard CSV and XLSX documents.	US_4	Upload CSV	Data Analyst	to upload a CSV file	I can load source data into the pipeline	As a Data Analyst, I want to upload a CSV file so that I can load my source data into the pipeline.	User can choose a CSV file from local storage. System validates the file format as CSV. System displays the loaded columns and row count.	High	Done
5	FR_4	The system shall allow users to import source data by uploading supported file formats, specifically including standard CSV and XLSX documents.	US_5	Upload XLSX	Data Analyst	to upload an XLSX file as a data source	I can use existing spreadsheet data in the pipeline	As a Data Analyst, I want to upload an XLSX file as a data source so that I can use existing spreadsheet data in the pipeline.	User selects an XLSX file from local storage. System validates the file format and reads the first sheet. Loaded columns and row count are displayed.	Medium	Done
6	FR_5	The system shall provide a canvas where users can add pipeline nodes.	US_6	Add nodes to canvas	Pipeline Builder	to drag nodes onto the canvas	I can visually build the data workflow	As a Pipeline Builder, I want to drag nodes onto the canvas so that I can build the data workflow visually.	Sidebar shows available node types. User can drag a node onto the canvas. Node appears on the canvas with a visible label.	High	Done
7	FR_6	The system shall allow users to connect nodes with edges to define data flow.	US_7	Connect nodes	Data Engineer	to connect node handles	the transformation order is defined correctly	As a Data Engineer, I want to connect nodes so that I can define the data flow order.	User can connect output handle to input handle. System creates a valid edge when connection is allowed. System prevents invalid connections and shows an error message.	High	To Do
8	FR_7	The system shall allow users to configure node-specific parameters.	US_8	Configure node settings	Analytics User	to open node configuration	I can specify transformation behavior	As an Analytics User, I want to configure node settings so that each transformation behaves as intended.	Clicking a node opens its configuration panel. User can change node-specific properties. Configuration changes are saved for the selected node.	High	To Do
9	FR_8	The system shall allow users to filter rows based on column conditions.	US_9	Filter rows	Data Analyst	to filter rows based on column conditions	I can isolate relevant records	As a Data Analyst, I want to filter rows based on column conditions so that I can isolate relevant records.	User selects column, operator, and value. System applies the filter and updates the preview. Rows not matching the condition are excluded from output.	High	In Progress
10	FR_9	The system shall allow users to sort rows by a selected column.	US_10	Sort data	Data Analyst	to sort rows by a selected column	I can order records for analysis	As a Data Analyst, I want to sort rows by a selected column so that I can order records for analysis.	User selects column and sort direction (ascending/descending). System reorders rows accordingly. Preview reflects the new row order.	Medium	In Progress
11	FR_10	The system shall allow users to select a subset of columns to keep.	US_11	Select columns	Pipeline Builder	to choose a subset of columns to keep	I can remove unnecessary data from the pipeline	As a Pipeline Builder, I want to choose a subset of columns to keep so that I can remove unnecessary data from the pipeline.	User selects columns to retain. System drops all other columns. Preview shows only the retained columns.	Medium	In Progress
12	FR_11	The system shall allow users to remove duplicate rows.	US_12	Remove duplicate rows	Data Analyst	to remove duplicate rows	my dataset contains only unique records	As a Data Analyst, I want to remove duplicate rows so that my dataset contains only unique records.	User selects column(s) to check for duplicates. System removes duplicate rows based on the selection. Preview shows the deduplicated row count.	Medium	In Progress
13	FR_12	The system shall allow users to fill missing values in a column.	US_13	Fill missing values	Data Analyst	to fill missing values in a column	incomplete data does not break downstream calculations	As a Data Analyst, I want to fill missing values in a column so that incomplete data does not break downstream calculations.	User selects a column and fill strategy (constant, mean, median, forward-fill). System replaces missing values accordingly. Preview shows the updated values.	Low	To Do
14	FR_13	The system shall allow users to add a new column computed from an expression.	US_14	Add calculated column	Analytics User	to add a new column computed from an expression	I can derive new metrics from existing data	As an Analytics User, I want to add a new column computed from an expression so that I can derive new metrics from existing data.	User enters a formula referencing existing columns. System validates the expression syntax. New column appears in the preview with computed values.	High	To Do
15	FR_14	The system shall allow users to group rows by a column, apply an aggregate function and join two data sources on a common key.	US_15	Group and aggregate data	Data Analyst	to group rows by a column and apply an aggregate function	I can summarize data	As a Data Analyst, I want to group rows by a column and apply an aggregate function so that I can summarize data.	User selects grouping column(s) and aggregate function (sum, count, average, min, max). System produces a grouped summary table. Preview reflects the aggregated results.	High	In Progress
16	FR_14	The system shall allow users to group rows by a column, apply an aggregate function and join two data sources on a common key.	US_16	Join datasets	Data Engineer	to join two data sources on a common key	I can combine related data into one dataset	As a Data Engineer, I want to join two data sources on a common key so that I can combine related data into one dataset.	User selects two input nodes and a join key. User selects join type (inner, left, right, outer). System produces the combined output.	High	To Do
17	FR_15	The system shall preview node output data for the selected node.	US_17	Preview node output	Reviewer	to preview node output	I can verify the transformation result	As a Reviewer, I want to preview node output so that I can verify the transformation result before full execution.	User can select a node and view output preview. Preview shows the first rows of node output. Preview updates after node configuration changes.	High	To Do
18	FR_16	The system shall prevent users from creating cyclic connections between nodes.	US_18	Validate pipeline structure	Data Engineer	the system to prevent cyclic node connections	the pipeline always represents a valid execution order	As a Data Engineer, I want the system to prevent cyclic node connections so that the pipeline always represents a valid execution order.	System checks for cycles when a new edge is created. System rejects connections that would create a cycle. User is shown an explanatory error message.	High	To Do
19	FR_17	The system shall visually highlight nodes that fail during execution.	US_19	Highlight faulty node	Reviewer	faulty nodes to be visually highlighted	I can quickly locate errors in the pipeline	As a Reviewer, I want faulty nodes to be visually highlighted so that I can quickly locate errors in the pipeline.	System detects node execution errors. Faulty node displays a red glow indicator. Error message is shown on hover or click.	High	To Do
20	FR_18	The system shall generate the equivalent Excel formula for a selected node.	US_20	Generate Excel formula	Spreadsheet User	to see the formula for a node	I can reproduce the logic in Excel	As a Spreadsheet User, I want to generate an Excel formula so that I can reproduce the pipeline logic in Excel.	System displays an Excel formula for the selected node. Formula reflects the node's configured logic. Formula is visible without requiring full pipeline execution.	Medium	To Do
21	FR_19	The system shall allow users to paste an Excel formula and have it converted into a visual pipeline.	US_21	Convert formula to pipeline	Spreadsheet User	to paste an Excel formula and have it converted to a visual pipeline	I can understand and reuse existing formula logic visually	As a Spreadsheet User, I want to paste an Excel formula and have it converted to a visual pipeline so that I can understand and reuse existing formula logic visually.	User pastes a formula into the input panel. System parses the formula into pipeline nodes. Nodes appear on the canvas with correct connections.	Medium	To Do
22	FR_20	The system shall execute the pipeline and display the final output.	US_22	Execute pipeline	Data Analyst	to run the pipeline	I can obtain the processed dataset	As a Data Analyst, I want to execute the pipeline so that I can obtain the processed dataset.	User can trigger pipeline execution. System processes the pipeline end to end. Final output or an execution error is shown to the user.	High	To Do
23	FR_21	The system shall provide a step-by-step animated replay of data flowing through the pipeline.	US_23	Replay pipeline execution	Reviewer	to play a step-by-step animation of data flowing through the pipeline	I can understand how data is transformed at each stage	As a Reviewer, I want to play a step-by-step animation of data flowing through the pipeline so that I can understand how data is transformed at each stage.	User clicks the replay button. System animates data flow sequentially through each connected node. User can pause and resume the animation.	Low	To Do
24	FR_22	The system shall allow registered users to save and reload pipeline templates.	US_24	Save pipeline template	Registered User	to save the current pipeline	I can reuse the workflow later	As a Registered User, I want to save the pipeline so that I can reuse it later.	User can save a pipeline as a template. Saved templates are listed in the template browser. User can reload a saved template onto the canvas.	Medium	To Do
25	FR_23	The system shall allow users to export the final pipeline output as a CSV file.	US_25	Export as CSV	Data Analyst	to export the final pipeline output as a CSV file	I can use the result outside the application	As a Data Analyst, I want to export the final pipeline output as a CSV file so that I can use the result outside the application.	User clicks export and selects CSV format. System generates a downloadable CSV file. File contents match the final preview.	High	To Do